ADR IDENTIFICATION
CASE STUDY

1. Patient Details

A 68-year-old woman weighing 58 kg had a history of hypertension and dyslipidemia. Her baseline serum creatinine was 0.9 mg/dL, and she had no previous muscle-related complaints.

Medication	Dose	Duration
Simvastatin	40 mg at bedtime (HS)	3 years
Amlodipine	5 mg once daily (OD)	3 years
Hydrochlorothiazide	12.5 mg once daily (OD)	2 years
Clarithromycin	500 mg twice daily (BD), prescribed for 7 days	Newly initiated on Day 0

Clarithromycin was prescribed for a lower respiratory tract infection. The prescriber was unaware that the patient was taking simvastatin.

2. Clinical Timeline

Day	Event
0	Clarithromycin started.
3	Mild thigh and calf pain developed; initially attributed to the infection.
4	Marked proximal muscle weakness, difficulty climbing stairs and cola-coloured urine.
6	Hospital admission. Clarithromycin and simvastatin discontinued.
6–10	Intravenous fluids and electrolyte monitoring.
10	Creatine kinase (CK) levels began trending down and urine colour cleared.
14	Creatinine improved to 1.0 mg/dL and muscle weakness resolved.

3. Admission Laboratory Findings (Day 6)

Parameter	Result	Reference range
Creatine kinase (CK)	18,500 U/L	Less than 170 U/L
Serum creatinine	1.9 mg/dL	0.5–1.1 mg/dL
Potassium	5.6 mmol/L	3.5–5.0 mmol/L
AST	420 U/L	Less than 40 U/L
ALT	210 U/L	Less than 40 U/L
Urine myoglobin	Positive	Negative
TSH	Normal	Within normal range

Clinical interpretation: The markedly elevated CK and positive urine myoglobin support severe skeletal muscle injury. The increased creatinine indicates acute kidney injury, while elevated

potassium is consistent with muscle-cell injury. AST and ALT may also rise due to skeletal muscle damage; their elevation alone does not establish primary liver injury.

Alternative causes including trauma, strenuous exercise, alcohol use and seizures were excluded. Normal TSH makes hypothyroidism less likely.

4. ADR Identification and Suspected Drugs

The presentation is strongly suggestive of rhabdomyolysis associated with a drug–drug interaction. Muscle symptoms began three to four days after clarithromycin was added, despite three years of prior simvastatin use without muscle complaints. The improvement after withdrawal of the suspected medicines supports a drug-related association.

Role	Drug	Assessment
Precipitating drug	Clarithromycin	Strong CYP3A4 inhibitor; can markedly increase exposure to simvastatin.
Object (affected) drug	Simvastatin	CYP3A4 substrate; increased exposure can cause severe muscle toxicity.
Contributing factor	Amlodipine	The recommended simvastatin maximum with amlodipine is 20 mg/day; the patient was taking 40 mg/day.
Not considered a primary suspect	Hydrochlorothiazide	Used for two years without prior muscle symptoms; no clear evidence implicates it in this event.

5. Mechanism of the Interaction

This is a pharmacokinetic drug–drug interaction. Clarithromycin strongly inhibits CYP3A4, an enzyme involved in simvastatin metabolism. Inhibition of this pathway increases simvastatin exposure and the risk of myotoxicity. Severe muscle-cell breakdown releases CK, myoglobin and potassium into the circulation. Myoglobin can contribute to renal injury, explaining the acute kidney injury observed in this patient.

The reaction is consistent with a predictable, dose-related Type A adverse drug reaction. The interaction is the key trigger; the pre-existing simvastatin dose above the recommended limit with amlodipine may have further increased the patient's risk.